UNIT 2 · REGRESSION ANALYSIS

Normal Linear Models

Maximum Likelihood Estimation, Coefficient Interpretation, and Inference in the Linear Model

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The normal model: a special case of GLM

GLM is a model where Y belongs to the exponential family.
When Y is normal, that particular case is what we call a **Normal Linear Model**

Generalized Linear Model

$$g(\mu_i) = x_i^T \beta$$

Y_i belongs to the exponential family

$g(.)$ is the link function



link = identity

Normal Linear Model

$$E(Y_i) = \mu_i = x_i^T \beta \;\; ; \;\; Y_i \sim N(\mu, \sigma^2)$$

$$y = X\beta + e \, , \;\; e_i \sim N(0, \sigma^2) \text{ i.i.d.}$$

The link function $g(\mu) = \mu$ doesn't transform anything: that's why the model looks "linear and direct," unlike logistic or Poisson.

Normal Linear Models

$$y = X\beta + e$$
$$Y \sim \text{Normal}$$

Multiple Regression

Primary objective: explain or predict a continuous outcome.

Predictors: can be continuous, categorical, or both.

Example: Predict monthly sales using advertising spend and store size.

ANOVA

Primary objective: compare the mean of a continuous outcome across groups.

Predictors: categorical variables only.

Example: Do average monthly sales differ across store formats?

ANCOVA

Primary objective: compare group means while adjusting for a continuous covariate.

Predictors: categorical variables + continuous covariates.

Example: Do average monthly sales differ across store formats after adjusting for advertising spend?

LINEAR REGRESSION

Simple vs. multiple regression

Simple

$$y = \beta_0 + \beta_1 x_1 + e$$

A single predictor explains the response variable.

Multiple

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_t x_t + e$$

Several predictors act simultaneously, each with its own coefficient.

Matrix notation (equivalent, for any number of predictors):

$$\mathbf{y = X\beta + e}$$

y : vector of n observations of the dependent variable

X : $n \times (p+1)$ matrix of predictors (includes a column of 1's for β_0)

β : vector of $p+1$ coefficients to estimate

e : vector of errors, $e_i \sim N(0, \sigma^2)$ independent

Continuous variables: the marginal change

$\beta_j = \Delta y / \Delta x_j$ Expected change in y for each additional unit of x_j, holding all other variables constant ("ceteris paribus").

Example with the banking credit dataset

For **each additional** month of job tenure (antiguedad_laboral_meses), the credit line amount **increases or decreases by β** pesos, holding everything else constant.

Careful with the intercept β_0

β_0 is the **expected value** of **y** when **ALL x's are zero.**
But does it make sense for a client with edad_cliente = 0 or ingreso_mensual = 0?
When **zero isn't an observable real value**, the **intercept loses practical interpretation**; it's still **mathematically necessary but isn't reported as a business finding.**

Reading Coefficients With Real Numbers

Same idea as before $\beta_j = \Delta y / \Delta x_j$ but now with real signs and magnitudes, so you can see β can push in either direction.

antiguedad_laboral_meses

β = +$150

↑ positive effect

For each **additional month of job tenure**, the **credit line amount increases** by **$150 pesos**, holding everything else constant.

score_buro

β = +$85

↑ positive effect

For each **additional point in credit bureau score**, the **credit line amount increases** by **$85 pesos**, holding everything else constant.

pagos_tardios_12m

β = −$620

↓ negative effect

For each **additional late payment in the last 12 months**, the **credit line amount decreases** by **$620 pesos**, holding everything else constant.

The **sign of β is never assumed** in advance, it's only known once the model is estimated from the data.

Categorical variables: why can't they go in directly?

The model $y = \beta_0 + \beta_1 x + e$ needs x to be numeric. "tipo_empleo" = "independiente" can't be multiplied by a coefficient.

✗ **Incorrect**

cliente_id	tipo_empleo
101	Empleado formal
102	Independiente
103	Empleado formal

There's no "β × Empleado formal" | text has no defined arithmetic operation.

✓ **Encoded (dummy encoding)**

cliente_id	tipo_empleo_independiente
101	0
102	1
103	0

Now yes: $y = \beta_0 + \beta_1(\text{tipo_empleo_independiente}) + e$ is a valid operation.

The coefficient of a dummy is a subtraction

Take the simplest case: $y = \beta_0 + \beta_1 D$, with $D \in \{0, 1\}$

These are two equations hidden inside one:

If D = 0 (ref.)

$$\hat{y} = \beta_0$$

-

If D = 1

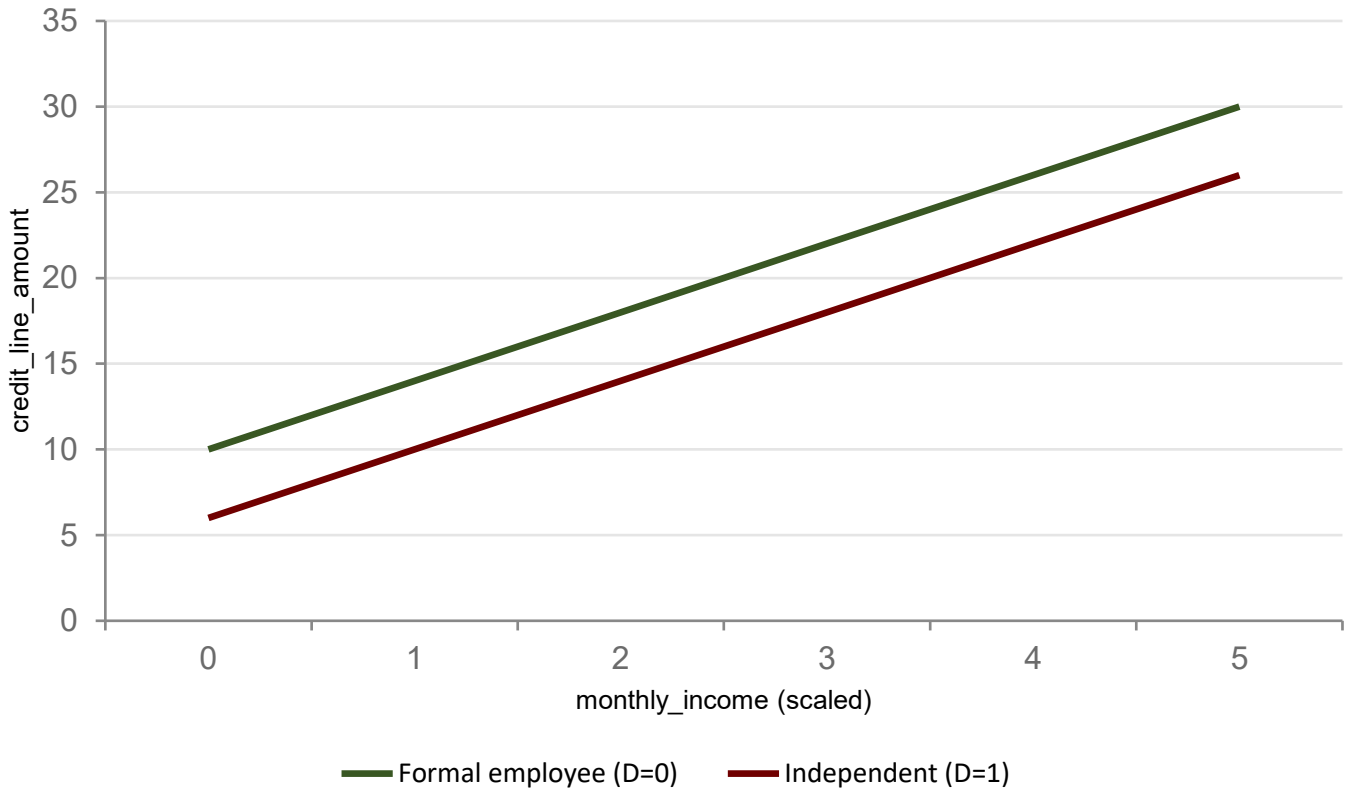
$$\hat{y} = \beta_0 + \beta_1$$

Subtracting the first equation from the second:

$$\beta_1 = (\beta_0 + \beta_1) - \beta_0 = \hat{y}(D=1) - \hat{y}(D=0)$$

β_1 is exactly the difference between the predicted value of one group and the other — the same "subtraction" logic you already used for continuous variables, except here the subtraction is between two groups, not between two units of x.

Visually: two parallel lines



Same slope, different intercept

Both lines rise by the same amount for every peso of income; the effect of x is the same in both groups.

The vertical distance between them is constant and is exactly β_1, the dummy's coefficient.

With more than 2 categories: several parallel lines, each shifted by its own coefficient.

Reference category and variables with k levels

General rule: k categories → k – 1 dummies

One category always stays out as the reference (baseline); its effect is "absorbed" into β_0. All others are interpreted relative to it.

Example: historial_pagos (Bueno / Regular / Malo) — reference: "Bueno"

historial_pagos	hist_Regular	hist_Malo
Bueno (reference)	0	0
Regular	1	0
Malo	0	1

Reading the coefficients:

β(hist_Regular) = expected difference in the credit amount between "Regular" and "Bueno," holding everything else constant.
β(hist_Malo) = the same, but against "Malo." There's no coefficient for "Bueno" — it's the yardstick.

Note: β_0 only equals the "Bueno" group's real average when historial_pagos is the model's only predictor. Once you add more variables (categorical or continuous), β_0 becomes a mathematical reference point — not a group you can observe in your data.

Reading Coefficients With Real Numbers

Same historial_pagos example — now reading the sign and magnitude of each coefficient against the reference group ("Bueno").

β_0 (Bueno — reference)

$\beta = \$45{,}000$

baseline group

The **average credit line for the reference group ("Bueno")**. Every other coefficient is read relative to this.

Note: this is a real group average only if **historial_pagos is the model's only predictor**

hist_Regular

$\beta = -\$2{,}500$

↓ negative effect

"Regular" clients get, on average, **$2,500 LESS** credit **than "Bueno" clients**, holding everything else constant.

hist_Malo

$\beta = -\$7{,}800$

↓ negative effect

"**Malo" clients** get, on average, $**7,800 LESS** credit **than "Bueno" clients**, holding everything else constant.

Comparing two non-reference categories:

β(hist_Malo) − β(hist_Regular) = −7,800 − (−2,500) = −$5,300
"Malo" clients get $5,300 less credit than "Regular" clients.

Careful: the size of β doesn't indicate importance

Comparing the size of two coefficients is only valid if their variables are on the same scale. With different scales, a large β may simply reflect small units.

Variable	Typical scale	Hypothetical β	"More important?"
ingreso_mensual	$8,000 – $80,000 MXN	0.15	*Not necessarily*
score_buro	300 – 850 points	42.30	*Not necessarily*

β(score_buro) = 42.30 is much larger than **β(ingreso_mensual) = 0.15**
but **score_buro** moves in **tens/hundreds of points**, while **ingreso_mensual** moves in **thousands of pesos**.
This is **not a fair comparison of "importance."**

Note: to compare relative importance across predictors, variables can be standardized (z-score) before fitting the model, producing standardized coefficients. We won't go deeper here — it's a topic to revisit later.

Is each coefficient significant? The t-test

$H_0 : \beta_j = 0$ (x_j doesn't contribute to the model, controlling for the other variables)

$H_1 : \beta_j \neq 0$ (x_j does have an effect on y)

Variable	Coefficient	p-value	Decision (α = 0.05)
ingreso_mensual	0.15	< 0.001	Rejects H_o — significant
score_buro	42.30	0.032	Rejects H_o — significant
edad_cliente	3.10	0.410	Does not reject H_o — not significant

p value ≥ 0.05 → Accept H_o | p value < 0.05 → Reject H_o

Does the whole model work? The F-test

$H_0 : \beta_1 = \beta_2 = \ldots = \beta_t = 0$ **(no variable helps — the model explains nothing)**

$H_1 : \text{at least one } \beta_j \neq 0$ **(the model, jointly, does explain variation in y)**

Why isn't it enough to look at the individual t-tests?

The **t-test** evaluates **each variable separately**.
The **F-test** evaluates the **joint predictive power of the whole model at once.** It's possible (though rare) to have individually non-significant variables but a significant overall F, especially when predictors are correlated.

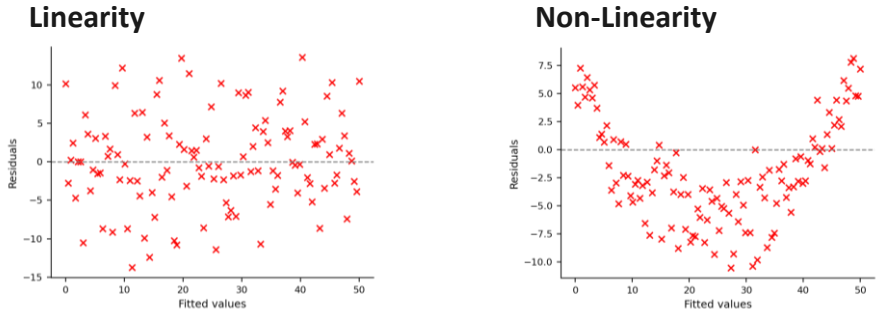
Example report (statsmodels)

F-statistic: 102.8 | Prob (F-statistic): 2.2e-16

p-value practically zero → we reject H_0: the model is useful for explaining the credit line amount.

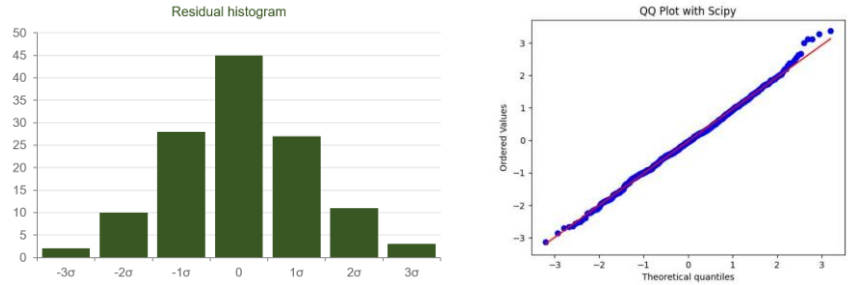
What Happens If an Assumption Is Violated?

1 **LINEARITY:** The relationship between X and Y is linear in the parameters.



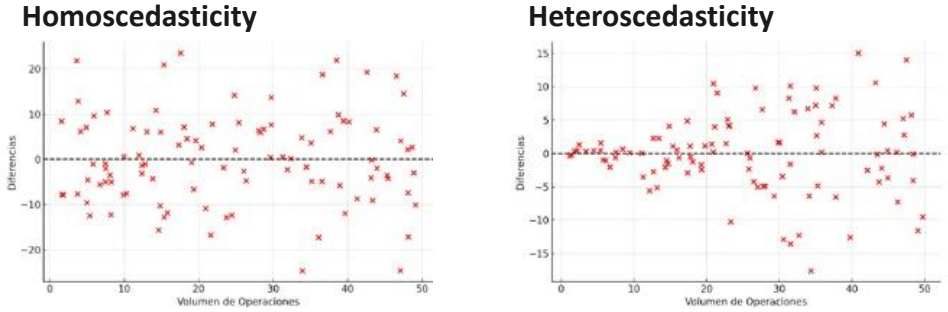
Linearity Test: Ramsey RESET

2 **NORMALITY:** The errors (and therefore Y) follow a normal distribution.



Normality Test: Shapiro-Wilk and Jarque-Bera

3 **HOMOSCEDASTICITY:** The variance of the error is constant across all levels of X.



Homoscedasticity Test: Breusch-Pagan

4 **INDEPENDENCE:** Observations and errors are independent of each other.

- **Independence:** each observation is unrelated to the others.
- **Non-independence:** one observation's error can be predicted from another's, usually from time order or grouping
- **Plot:** only possible when a time/grouping variable exists to plot against

Idependency Test: Durbin-Watson

If an Assumption Doesn’t Hold…

Recognizing which of two problems you're facing:

⚠ Inconsistent model behavior

The model itself is wrong in some way , it fits well for some ranges or groups of clients, and poorly for others. This happens mainly when linearity fails.

Example: the model predicts credit lines well for average incomes but is systematically wrong for very high or very low incomes.

⚠ Unreliable inference

The coefficients themselves are still fine, but you can no longer trust the p-values, confidence intervals, or significance tests. This happens when normality, homoscedasticity, or independence fail.

Example: you might conclude a variable is significant (or not) when that conclusion is unreliable.

R^2 and Adjusted R^2

R^2

Proportion of the variance in y explained by the model.

Problem: it ALWAYS goes up (or stays the same) when adding more variables, even irrelevant ones.

The **model explains 72% of the variation** in **monto_linea_credito** across clients.
The remaining 28% is due to factors the model doesn't capture (noise, missing variables, etc.).

Adjusted R^2

Same as R^2 but penalizes adding variables that don't add real explanatory power.

It's the right metric to compare models with a different number of predictors

Example 1 - Small gap
$R^2 = 0.72$R2=0.72, $R^2_{adjusted} = 0.70$

The two numbers are close, so the variables in the model are earning their place — this is a healthy model."

Example 2 - Large gap
$R^2 = 0.75$R2=0.75, $R^2_{adjusted} = 0.55$

The big drop from 0.75 to 0.55 means the model likely has variables that aren't really helping, just noise.

MAE, MAPE, and RMSE

MAE

Mean Absolute Error

$$\mathbf{MAE} = \frac{1}{n}\sum |\mathbf{y_i} - \hat{\mathbf{y}}_\mathbf{i}|$$

If $\mathbf{MAE} = \$3,200$:

On average, the model's prediction is off by $3,200 pesos from the actual credit line amount, regardless of whether it overshoots or undershoots.

MAPE

Mean Absolute Percentage Error

$$\mathbf{MAPE} = \frac{1}{n}\sum \frac{|\mathbf{y_i} - \hat{\mathbf{y}}_\mathbf{i}|}{|\mathbf{y_i}|}$$

If $\mathbf{MAPE} = 8.5\%$:

On average, the model's prediction is off by 8.5% of each client's actual credit amount.

Unlike MAE, this is comparable across clients with very different amounts a $3,200 peso error is huge for a client with a $10,000 credit line, but small for one with a $200,000 line.

RSME

Root Mean Squared Error

$$\mathbf{RSME} = \sqrt{\frac{1}{n}\sum(\mathbf{y_i} - \hat{\mathbf{y}}_\mathbf{i})^2}$$

Example — RMSE close to MAE

$\boldsymbol{MAE = \$2,800, RMSE = \$3,100}$
RMSE is only slightly higher than MAE. The model's errors are fairly consistent across clients.

Example — RMSE much higher than MAE

$\mathbf{MAE = \$2,800, RMSE = \$6,500}$
RMSE is much higher than MAE.
This gap signals a few large errors that are pulling the average up.

From theory to code

Everything we saw today, we’re going to apply on the same dataset you already know from the EDA:

credito_bancario.csv — monto_linea_credito ~ ingreso_mensual + score_buro + …

- Encoding categorical variables (tipo_empleo, estado_civil, historial_pagos)
- Train / test split
- Model 1 (simple) and Model 2 (multiple)
- Interpretation, residual diagnostics, and performance metrics
- Final comparison table